SUMESH MOHAN

DevOps Engineer & Autonomous AI Systems Architect

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Professional Profile:
Passionate DevOps Engineer & Autonomous AI Systems Architect with 4+ years of hands-on experience building resilient CI/CD pipelines, administering hybrid multi-cloud infrastructure (Azure & AWS), and deploying containerized applications with Docker and Jenkins. Specialized in architecting local AI agents (Ollama & Llama 3), building 24/7 cloud microservices, and sustaining 99.9% high-availability production environments with Zabbix observability.

TECHNICAL SKILLS & CORE COMPETENCIES

Cloud & Virtualization	Microsoft Azure (AZ-104) | Amazon Web Services (AWS) | VMware ESXi
CI/CD & DevOps Tools	Jenkins | Docker | Git | GitLab CI/CD | Pipeline Automation
Scripting & Automation	Python | Bash Shell | PowerShell | Automated Patching & Backup Scheduling
System Administration	Linux (Ubuntu, Debian, RHEL) | Windows Server | Active Directory | Disaster Recovery
AI Systems & Telemetry	Local LLM Orchestration (Ollama) | Neural Voice (Edge-TTS) | Zabbix | MySQL | MongoDB

PROFESSIONAL EXPERIENCE

DevOps Engineer
Jan 2022 - Present

PanApps Inc. (Assyst International) | Muttom, Aluva, Ernakulam, Kerala

- **Automated CI/CD Delivery:** Designed, hardened, and maintained multi-stage Jenkins CI/CD deployment pipelines, reducing build failure rates and accelerating enterprise software release frequency by 30%.
- **Cloud Infrastructure Optimization:** Provisioned, configured, and governed scalable cloud instances, virtual networks, and blob storage across Microsoft Azure and AWS, achieving optimized compute allocation and reduced operational overhead.
- **Containerization:** Built, optimized, and maintained lightweight Docker container images for microservices and web applications, ensuring consistent execution across dev, test, and production stages.
- **Scripted Automation:** Authored robust Python, Bash, and PowerShell automation scripts to handle unattended operating system updates, server patching, and scheduled data backups, reclaiming 15+ hours of manual labor per month.
- **Real-Time Observability:** Deployed and configured Zabbix monitoring agents across distributed server fleets, implementing custom trigger thresholds and proactive alerting that lowered incident response time by 22%.
- **High-Availability Systems:** Managed enterprise Windows Server and Linux environments with automated snapshots, disaster recovery protocols, and POSIX permission auditing, sustaining 99.9% SLA uptime.
- **Agile Collaboration & Version Control:** Established Git branching workflows, pull request governance, and merge guidelines supporting simultaneous parallel feature development across multiple agile development teams.
- **Web Infrastructure & Office 365:** Administered multi-tenant web application hosting, SSL/TLS certificate renewals, reverse proxies, and Microsoft 365 user lifecycle management and security policies.

Network Technician - Fiber Optics
Aug 2021 - Jan 2022

Reliance Jio Infocomm Ltd. | Kollam, Kerala

- **Network Deployment & Auditing:** Conducted end-to-end technical site feasibility surveys and customer infrastructure inspections to validate fiber delivery specifications and bandwidth requirements.
- **Hardware Configuration:** Installed, provisioned, and verified Optical Network Terminals (ONT), enterprise wireless access points, and IP set-top devices adhering strictly to telecommunication quality standards.
- **Optical Fiber Systems:** Executed precision fusion splicing and collaborated with departmental network engineering teams in deploying, testing, and fine-tuning laser transmission equipment.
- **Testing & Standards Compliance:** Performed optical signal loss testing, OTDR measurements, and equipment calibration according to manufacturer guidelines to guarantee link stability and peak transmission performance.

KEY TECHNICAL PROJECTS & AUTONOMOUS AI SYSTEMS

FRIDAY - Autonomous AI Desktop Agent & Voice Automation Engine — Python | Ollama (Llama 3.2) | Edge-TTS | PowerShell | Obsidian

- **Local AI & Voice Interaction:** Architected a fully voice-enabled autonomous desktop AI agent running entirely locally on Windows 11 with Ollama (Llama 3.2) and Microsoft Edge-TTS neural speech synthesis.
- **System Automation:** Implemented real-time PowerShell execution, unattended desktop workflow orchestration, multi-turn tool calling, and automated integration with an Obsidian persistent knowledge base (Karpathy LLM-Wiki).

Tech World News - 24/7 Cloud Intelligence Daemon & Webhook Gateway — Python | Docker | Render Cloud | Gemini Flash API | Telegram Bot API

- **24/7 Cloud Architecture:** Engineered a containerized news intelligence daemon on Render Cloud utilizing Telegram Webhooks and an automated 9-minute self-keepalive background loop to prevent idle sleep.
- **AI Pipeline & Tri-Layer Sync:** Integrated Google Gemini Flash API with multi-model fallback for objective, hype-free tech analysis, paired with direct mobile markdown delivery and automated GitHub repository commits.

Interactive 3D Datacenter Infrastructure & CI/CD Release Simulator — Three.js | WebGL | JavaScript | HTML5 / CSS3

- **3D Graphics & Physics:** Developed an interactive 3D web platform featuring a 12-rack enterprise datacenter corridor, scroll-driven server zoom, dynamic liquid ticker reel, and kinetic scroll-settling typography.
- **DevOps Simulator:** Built an interactive 5-stage CI/CD pipeline simulator modeling Git branching, Jenkins builds, multi-stage Docker builds, and Zabbix telemetry, alongside an operator CLI terminal emulator.

PROFESSIONAL CERTIFICATIONS

Microsoft Certified: Azure Administrator Associate (AZ-104) — Sep 2023 - Sep 2027

Demonstrates verified expertise in architecting, managing, and monitoring Azure cloud environments. Highly proficient in configuring VMs, virtual networks, blob storage, load balancers, Entra ID, RBAC security, and PowerShell/CLI automation.

Cisco Certified Network Associate (CCNA) — Jul 2021 - Jul 2024

Globally recognized certification validating core networking mastery: IPv4/IPv6 routing, VLAN segmentation, switching architectures, ACL security, NAT/DHCP, and enterprise WAN connectivity. Strong expertise in packet diagnostics and troubleshooting.

Diploma in Computer Applications (DCA) — Apr 2018 - Completed

Comprehensive program covering software engineering fundamentals, relational databases (SQL), programming in C, C++, Java, web development (HTML, CSS), and office productivity automation suites.

EDUCATION

Sree Narayana Polytechnic College | Kottiyam, Kollam — Jun 2017 - Apr 2020

Diploma in Electronics Engineering - 82% Aggregate

LANGUAGES & PROFESSIONAL STRENGTHS

LANGUAGES KNOWN

English: Professional Working Proficiency
Malayalam: Native / Mother Tongue
Tamil: Professional Working Proficiency

PROFESSIONAL ATTRIBUTES

- Analytical Problem-Solving & Root Cause Analysis
- Agile & Scrum Cross-Functional Team Collaboration
- High Adaptability in Fast-Paced Technical Environments
- Strict Commitment to 99.9% Uptime & Infrastructure Security